

Linux系统iptables防火墙设置

笔记本： linux加固

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使用yum来安装，安装命令如下：

```
1 | yum install -y iptables
```

iptables安装好后就可以来配置规则了。由于作为web服务器来使用，所以对外要开放 80 端口，另外肯定要通过ssh进行服务器管理，22 端口也要对外开放，当然最好是把ssh服务的默认端口改掉，在公网上会有很多人试图破解密码的，如果修改端口，记得要把该端口对外开发，否则连不上就悲剧了。下面提供配置规则的详细说明：

第一步：清空所有规则

当Chain INPUT (policy DROP)时执行/sbin/iptables -F后，你将和服务器断开连接
所有在清空所有规则前把policy DROP该为INPUT，防止悲剧发生，小心小心再小心

```
/sbin/iptables -F INPUT ACCEPT
```

清空所有规则

```
/sbin/iptables -F
```

```
/sbin/iptables -X
```

计数器置0

```
/sbin/iptables -Z
```

第二步：设置规则

允许来自于lo接口的数据包，如果没有此规则，你将不能通过127.0.0.1访问本地服务，例如

```
/sbin/iptables -A INPUT -i lo -j ACCEPT
```

开放TCP协议22端口，以便能ssh，如果你是在有固定ip的场所，可以使用 -s 来限定客户端

```
/sbin/iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

开放TCP协议80端口供web服务

```
/sbin/iptables -A INPUT -p tcp --dport 80 -j ACCEPT
```

10.241.121.15是另外一台服务器的内网ip，由于之间有通信，接受所有来自10.241.121.1

```
/sbin/iptables -A INPUT -p tcp -s 10.241.121.15 -j ACCEPT
```

接受ping

```
/sbin/iptables -A INPUT -p icmp -m icmp --icmp-type 8 -j ACCEPT
```

这条规则参看：<http://www.netingcn.com/iptables-localhost-not-access-interne>

```
/sbin/iptables -A INPUT -m state --state ESTABLISHED -j ACCEPT
```

屏蔽上述规则以为的所有请求，不可缺少，否则防火墙没有任何过滤的功能

```
/sbin/iptables -P INPUT DROP
```

可以使用 `iptables -L -n` 查看规则是否生效

至此防火墙就算配置好，但是这是临时的，当重启iptables或重启机器，上述配置就会被清空，要想永久生效，还需要如下操作：

```
1 /etc/init.d/iptables save
2 或
3 service iptables save
4
5 执行上述命令可以在文件 /etc/sysconfig/iptables 中看到配置
```

以下提供一个干净的配置脚本：

```
1 /sbin/iptables -P INPUT ACCEPT
2 /sbin/iptables -F
3 /sbin/iptables -X
4 /sbin/iptables -Z
5
6
7 /sbin/iptables -A INPUT -i lo -j ACCEPT
8 /sbin/iptables -A INPUT -p tcp --dport 22 -j ACCEPT
9 /sbin/iptables -A INPUT -p tcp --dport 80 -j ACCEPT
10 /sbin/iptables -A INPUT -p tcp -s 10.241.121.15 -j ACCEPT
11 /sbin/iptables -A INPUT -p icmp -m icmp --icmp-type 8 -j ACCEPT
12 /sbin/iptables -A INPUT -m state --state ESTABLISHED -j ACCEPT
/sbin/iptables -P INPUT DROP
```

最后执行 `service iptables save`，先确保ssh连接没有问题，防止规则错误，导致无法连上服务器，因为没有save，重启服务器规则都失效，否则就只有去机房才能修改规则了。也可以参考：[ubuntu iptables 配置脚本](#)来写一个脚本。

最后再次提醒，在清空规则之前一定要小心，确保Chain INPUT (policy ACCEPT)。

脚本之家补充阿里云的linux_drop_port.sh

```
1 #!/bin/bash
2 #####
3 #Function: linux drop port
4 #Usage: bash linux_drop_port.sh
5 #Author: Customer Service Department
6 #Company: Alibaba Cloud Computing
7 #Version: 2.0
8 #####
9
10 check_os_release()
11 {
12     while true
```

```
do
os_release=$(grep "Red Hat Enterprise Linux Server release"/etc/issue 2>
os_release_2=$(grep "Red Hat Enterprise Linux Server release"/etc/redhat
if [ "$os_release" ] && [ "$os_release_2" ]
then
    if echo "$os_release"|grep "release 5" >/dev/null2>&1
    then
        os_release=redhat5
        echo "$os_release"
    elif echo "$os_release"|grep "release 6">/dev/null 2>&1
    then
        os_release=redhat6
        echo "$os_release"
    else
        os_release=""
        echo "$os_release"
    fi
    break
fi
os_release=$(grep "Aliyun Linux release" /etc/issue2>/dev/null)
os_release_2=$(grep "Aliyun Linux release" /etc/aliyun-release2>/dev/nul
if [ "$os_release" ] && [ "$os_release_2" ]
then
    if echo "$os_release"|grep "release 5" >/dev/null2>&1
    then
        os_release=aliyun5
        echo "$os_release"
    elif echo "$os_release"|grep "release 6">/dev/null 2>&1
    then
        os_release=aliyun6
        echo "$os_release"
    else
        os_release=""
        echo "$os_release"
    fi
    break
fi
os_release=$(grep "CentOS release" /etc/issue 2>/dev/null)
os_release_2=$(grep "CentOS release" /etc/*release2>/dev/null)
if [ "$os_release" ] && [ "$os_release_2" ]
then
    if echo "$os_release"|grep "release 5" >/dev/null2>&1
    then
        os_release=centos5
        echo "$os_release"
    elif echo "$os_release"|grep "release 6">/dev/null 2>&1
    then
        os_release=centos6
        echo "$os_release"
```

```

82     else
83     os_release=""
84     echo "$os_release"
85     fi
86     break
87 fi
88
89 os_release=$(grep -i "ubuntu" /etc/issue 2>/dev/null)
90 os_release_2=$(grep -i "ubuntu" /etc/lsb-release2>/dev/null)
91 if [ "$os_release" ] && [ "$os_release_2" ]
92 then
93     if echo "$os_release"|grep "Ubuntu 10" >/dev/null2>&1
94     then
95         os_release=ubuntu10
96         echo "$os_release"
97     elif echo "$os_release"|grep "Ubuntu 12.04">/dev/null 2>&1
98     then
99         os_release=ubuntu1204
100        echo "$os_release"
101    elif echo "$os_release"|grep "Ubuntu 12.10">/dev/null 2>&1
102    then
103        os_release=ubuntu1210
104        echo "$os_release"
105    else
106        os_release=""
107        echo "$os_release"
108    fi
109    break
110 fi
111
112 os_release=$(grep -i "debian" /etc/issue 2>/dev/null)
113 os_release_2=$(grep -i "debian" /proc/version 2>/dev/null)
114 if [ "$os_release" ] && [ "$os_release_2" ]
115 then
116     if echo "$os_release"|grep "Linux 6" >/dev/null2>&1
117     then
118         os_release=debian6
119         echo "$os_release"
120     else
121         os_release=""
122         echo "$os_release"
123     fi
124     break
125 fi
126
127 os_release=$(grep "openSUSE" /etc/issue 2>/dev/null)
128 os_release_2=$(grep "openSUSE" /etc/*release 2>/dev/null)
129 if [ "$os_release" ] && [ "$os_release_2" ]
130 then
131     if echo "$os_release"|grep "13.1" >/dev/null 2>&1
132     then
133         os_release=opensuse131

```

```

147     echo "$os_release"
148 else
149     os_release=""
150     echo "$os_release"
151 fi
152 break
153 fi
154 break
155 done
156 }
157
158 exit_script()
159 {
160     echo -e "\033[1;40;31mInstall $1 error,will exit.\n\033[0m"
161     rm-f $LOCKfile
162     exit 1
163 }
164
165 config_iptables()
166 {
167     iptables -I OUTPUT 1 -p tcp -m multiport --dport 21,22,23,25,53,80,135,13
168     iptables -I OUTPUT 2 -p tcp -m multiport --dport 1433,1314,1521,2222,330
169     iptables -I OUTPUT 3 -p udp -j DROP
170     iptables -nvL
171 }
172
173 ubuntu_config_ufw()
174 {
175     ufwdeny out proto tcp to any port 21,22,23,25,53,80,135,139,443,445
176     ufwdeny out proto tcp to any port 1433,1314,1521,2222,3306,3433,3389,489
177     ufwdeny out proto udp to any
178     ufwstatus
179 }
180
181 #####Start#####
182 #check lock file ,one time only let thescript run one time
183 LOCKfile=/tmp/.$(basename $0)
184 if [ -f "$LOCKfile" ]
185 then
186     echo -e "\033[1;40;31mThe script is already exist,please next timeto run
187     exit
188 else
189     echo -e "\033[40;32mStep 1.No lock file,begin to create lock fileand con
190     touch $LOCKfile
191 fi
192
193 #check user
194 if [ $(id -u) != "0" ]
195 then

```

```

echo -e "\033[1;40;31mError: You must be root to run this script,please
rm-f $LOCKfile
exit 1
fi

echo -e "\033[40;32mStep 2.Begen tocheck the OS issue.\n\033[40;37m"
os_release=$(check_os_release)
if [ "X$os_release" == "X" ]
then
    echo -e "\033[1;40;31mThe OS does not identify,So this script isnot exec
rm-f $LOCKfile
exit 0
else
    echo -e "\033[40;32mThis OS is $os_release.\n\033[40;37m"
fi

echo -e "\033[40;32mStep 3.Begen toconfig firewall.\n\033[40;37m"
case "$os_release" in
redhat5|centos5|redhat6|centos6|aliyun5|aliyun6)
    service iptables start
    config_iptables
    ;;
debian6)
    config_iptables
    ;;
ubuntu10|ubuntu1204|ubuntu1210)
    ufwenable <<EOF
y
EOF
    ubuntu_config_ufw
    ;;
opensuse131)
    config_iptables
    ;;
esac

echo -e "\033[40;32mConfig firewallsuccess,this script now exit!\n\033[40
rm -f $LOCKfile

```